

Counterexamples to weak chromatic splitting: sphere kernels and descent exponents

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Abstract

For the derived p -completion of the sphere, the canonical weak chromatic splitting map has no homotopy retraction at height p for every prime $p \geq 5$, and at height $p + 1$ for every prime $p \geq 7$. The classical sphere product $\beta_1^{(p-1)^2}$ gives a nonzero kernel class in both ranges.

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1 The canonical weak splitting map

Fix a prime p and work in ordinary p -local spectra, with sphere $S = S_{(p)}^0$. Let $K(a)$ denote Morava K -theory of height a , put $K(0) = H\mathbb{Q}$, and write

$$L_r = L_{K(0) \vee \dots \vee K(r)}, \quad X = S_p^\wedge = \operatorname{holim}_{j \geq 1} S/p^j.$$

Thus L_r is Johnson–Wilson $E(r)$ -localization. For a spectrum Z , let $\eta_Z^{(n)} : Z \rightarrow L_{K(n)}Z$ be the localization unit. We study

$$i_{n,X} = L_{n-1}(\eta_X^{(n)}) : L_{n-1}X \longrightarrow L_{n-1}L_{K(n)}X. \quad (1)$$

The weak chromatic splitting conjecture predicts a homotopy retraction of this map whenever X is the derived p -completion of a finite spectrum. A retraction is a map of spectra r with $ri_{n,X} \simeq \operatorname{id}$; it need not be natural in the finite input. We disprove this universal assertion with finite input the sphere.

For an odd prime, let

$$\beta_1 \in \pi_{d_p}S, \quad d_p = 2p(p-1) - 2, \quad k_p = (p-1)^2, \quad z_p = \beta_1^{k_p},$$

where β_1 is the classical first beta element.

Theorem 1.1 (A named kernel at height p). *For every prime $p \geq 5$, the image of z_p in $\pi_{d_p k_p}L_{p-1}X$ is nonzero and is killed by $\pi_{d_p k_p}(i_{p,X})$. Consequently $i_{p,X}$ has no homotopy retraction.*

Theorem 1.2 (The same kernel at height $p+1$). *For every prime $p \geq 7$, the image of the same z_p in $\pi_{d_p k_p}L_pX$ is nonzero and is killed by $\pi_{d_p k_p}(i_{p+1,X})$. Consequently $i_{p+1,X}$ has no homotopy retraction.*

Corollary 1.3 (The prime-five class). *At $p = 5$, the image of β_1^{16} in $\pi_{608}L_4X$ is nonzero and belongs to the kernel of $\pi_{608}(i_{5,X})$.*

The proof detects z_p under a height- $(p-1)$ unit and kills it under both upper units. The lower detection follows from the Hopkins–Miller calculation recorded by Heard [10, proof of Theorem 5.6 and Lemma 5.8]: the actual unit detects the named β_1 , and its k_p th power remains nonzero in the ordinary homotopy fixed point spectral sequence. At the upper height a , we construct a separated multiplicative filtration with the unit image of β_1 in filtration at least two and $F^{a^2+2} = 0$. The required inequalities are

$$2k_p - (p^2 + 1) = p^2 - 4p + 1 > 0 \quad (p \geq 5),$$

$$2k_p - ((p+1)^2 + 1) = p(p-6) > 0 \quad (p \geq 7).$$

Completion and naturality then place the nonzero lower image in the kernel of the exact map (1). For example, at $p = 7$ the class β_1^{36} has degree $82 \cdot 36 = 2952$ and gives kernels at both heights seven and eight.

Notation. Let E_a be Morava E -theory of the height- a Honda formal group over $\mathbb{F}_{p^a}$, and put $B_a = L_{K(a)}S$. Unless a localized tensor is explicitly displayed, smash products and function spectra are ordinary. All homotopy groups are those of the underlying spectra.

1.1 An independent descent-exponent obstruction

The height- p nonretraction also follows from incompatible finite descent lengths. Put $t = p - 1$, and let the finite pure-stabilizer subgroup $H = C_p \rtimes C_{(p-1)^2}$ act on E_t . In $\mathcal{D}_H = \text{Fun}(BH, \text{Sp}_{(p)})$, with pointwise ordinary smash, put $A_H = F(H_+, S)$ with its translation action and $J_H = \text{fib}(S \rightarrow A_H)$. Define $\exp_H(V)$ as the least $b \geq 1$ such that $J_H^{\wedge b} \wedge V \rightarrow V$ is null, and as infinity if no such b exists. Set

$$T(V) = L_{K(t)}(E_t \wedge V),$$

with H acting on the first factor.

Theorem 1.4 (Incompatible descent exponents). *For every prime $p \geq 5$, with $m = p^2 + 2$,*

$$\exp_H(E_{p-1}) > m, \quad \exp_H(T(B_p)) \leq m.$$

A homotopy retraction of $i_{p,X}$ would make E_{p-1} an H -equivariant retract of $T(B_p)$. Hence $i_{p,X}$ has no homotopy retraction, independently of Theorem 1.1.

For the lower bound, a class survives to one specified finite page of the ordinary homotopy fixed point spectral sequence. No permanent cycle or sphere representative is needed. For the upper bound, Devinatz–Hopkins’ nilpotence theorem supplies a finite exponent before any homotopy test is chosen [7, Appendix A, Proposition A.3]. The numerical cochain bound then makes the relevant map zero after every finite ordinary test. A Brown–Comenetz argument proves that the map itself is null, and a coherent trace construction transports the resulting finite filtration to height $p - 1$. The finite-stage formalism used here belongs to the nilpotence and descent theory of Mathew and Mathew–Naumann–Noel [20, 21, 22].

1.2 Relation to earlier splitting results

Hopkins’ chromatic splitting conjecture, recorded by Hovey [13, Conjecture 4.2, especially part (v)], predicts a specific decomposition of chromatic overlap as well as the weak unit retraction. These predictions differ already at $n = p = 2$. Beaudry disproves the proposed strong summand formula [3, Theorem 1.0.4]; Beaudry–Goerss–Henn give a revised decomposition with Moore terms that retains the canonical unit as a retract [4, Theorem 1.1.6]. Their Conjectures 1.1.10–1.1.11 distinguish the strong and completed-finite-input weak formulations, and their introduction records the weak form at all primes through height two. The theorem numbers refer to the versioned texts identified in the bibliography. Outside finite inputs, Minami reports Devinatz’s counterexample for a BP analogue allowing completed BP [23, Introduction and Remark 1.1]. The present counterexamples use the completion of the finite sphere.

Long beta powers in the sphere were known earlier. Lee–Ravenel prove $\beta_1^{p^2-p-1} \neq 0$ for $p \geq 7$ and report $\beta_1^{17} \neq 0$, $\beta_1^{18} = 0$ at five [18, opening theorem and introduction]. Ravenel’s first-beta detection calculation relates the cobar class to stabilizer cohomology [25, Theorem 4 and its proof in Section 2]. The additional requirement here is detection after a specified lower localization and vanishing of that same named sphere product under the upper unit.

Organization. Sections 2–5 prove the named kernel results. Their upper bound uses dimension at height $p + 1$ and a semilinear trace-one contraction at height p , retaining the possible inverse-limit degree in both cases. Sections 6–11 give the independent exponent proof. Appendix A proves the ordinary filtered-product lemma used by the lower detector and by the finite-page argument.

2 Completion and the canonical map

The lower detectors are local spectra, whereas the weak splitting problem uses a completed finite input. The following argument connects these settings without interchanging smash products and inverse limits.

Lemma 2.1. *The completion map $c : S \rightarrow X = S_p^\wedge$ is a $K(a)$ -equivalence for every integer $a \geq 1$. If $1 \leq t \leq r$ and Y is $K(t)$ -local, every map $S \rightarrow Y$ factors through $S \rightarrow X \rightarrow L_r X$.*

Proof. Take the inverse limit of the cofiber sequences $S \xrightarrow{p^j} S \rightarrow S/p^j$, with multiplication by p on the left spheres and identity on the middle spheres. The fiber of c is

$$C \simeq \operatorname{holim}(\cdots \xrightarrow{p} S \xrightarrow{p} S) \simeq F(S[1/p], S).$$

Multiplication by p is an equivalence on this function spectrum, because it is an equivalence on its first argument. It remains an equivalence after ordinary smash with $K(a)$. But $\pi_*(K(a) \wedge C)$ is a module over $K(a)_*$ and is annihilated by p . Thus it is zero, proving the first claim.

A map to a $K(t)$ -local target consequently extends across c . The target is also L_r -local when $t \leq r$, since an L_r -acyclic spectrum is $K(t)$ -acyclic. The extension therefore factors through $L_r X$. $\square$

Proposition 2.2 (An actual sphere kernel). *Let $n \geq 2$, let $1 \leq t \leq n-1$, and let $z \in \pi_q S$. Suppose a map $S \rightarrow Y$ to a $K(t)$ -local spectrum carries z to a nonzero element, and the unit $S \rightarrow L_{K(n)} S$ carries z to zero. Then the image $\bar{z} \in \pi_q L_{n-1} X$ is nonzero and $\pi_q(i_{n,X})(\bar{z}) = 0$.*

Proof. Lemma 2.1 factors the detecting map through $L_{n-1} X$, so $\bar{z}$ is nonzero. Write η for $K(n)$ -localization and λ for L_{n-1} -localization. The relevant two squares are

$$\begin{array}{ccccc} S & \xrightarrow{c} & X & & X & \xrightarrow{\eta_X} & L_{K(n)} X \\ \eta_S \downarrow & & \downarrow \eta_X & & \lambda_X \downarrow & & \downarrow \lambda_{L_{K(n)} X} \\ L_{K(n)} S & \xrightarrow{L_{K(n)} c} & L_{K(n)} X & & L_{n-1} X & \xrightarrow{i_{n,X}} & L_{n-1} L_{K(n)} X. \end{array}$$

Both commute by naturality of localization units. Equivalently,

$$\eta_X c \simeq L_{K(n)}(c) \eta_S, \quad i_{n,X} \lambda_X \simeq \lambda_{L_{K(n)} X} \eta_X.$$

The first sends z to zero, and the second then gives the asserted kernel of the exact map (1). A retraction would induce a left inverse on this homotopy group and is therefore impossible. $\square$

3 The named sphere product at the lower height

The classical first beta element supplies the lower detector at every odd prime. The fixed-point calculation identifies its image under an actual unit, and the ordinary multiplicative filtration then detects its long power. This gives one named sphere product for both upper heights.

Fix an odd prime p and put

$$h = p - 1, \quad d = 2p(p - 1) - 2, \quad z_p = \beta_1^{(p-1)^2}, \quad \beta_1 \in \pi_d S.$$

Theorem 3.1 (The lower sphere product). *The image of z_p is nonzero in each of*

$$\pi_{d(p-1)^2} L_{K(p-1)} S, \quad \pi_{d(p-1)^2} L_{K(p-1)} X, \quad \pi_{d(p-1)^2} L_r X \quad (r \geq p - 1).$$

All three images are induced by the sphere unit and, for the completed input, by the completion map $S \rightarrow X$.

3.1 The actual unit in the ordinary detector

Let E_h be Morava E -theory over $\mathbb{F}_{p^h}$. Write $\mathbb{S}_h$ for the pure stabilizer and $\mathbb{G}_h = \mathbb{S}_h \rtimes \text{Gal}(\mathbb{F}_{p^h}/\mathbb{F}_p)$ for the extended stabilizer. Choose the finite subgroup $G \subset \mathbb{G}_h$ whose intersection with the pure stabilizer is $C_p \rtimes C_{h^2}$ and whose Galois projection is onto, as in [10, Section 2]. Put

$$Y = E_h^{hG}, \quad \eta : S \longrightarrow Y.$$

Here Y is the ordinary homotopy fixed point spectrum and η is its unit as a commutative ring spectrum. Devinatz–Hopkins identify the ordinary finite-group fixed-point spectral sequence with the local Adams spectral sequence and give strong convergence [7, Theorems 2(ii) and 3]. Its decreasing homotopy filtration $F^s \pi_* Y$ is therefore separated, with

$$F^s \pi_q Y / F^{s+1} \pi_q Y \cong E_\infty^{s, q+s}.$$

The Hopkins–Miller calculation gives the positive-filtration part of its E_2 page as the corresponding part of

$$\mathbb{F}_p[\alpha, \beta, \Delta^{\pm 1}] / (\alpha^2), \quad |\alpha| = (1, 2h), \quad |\beta| = (2, 2ph), \quad |\Delta| = (0, 2ph^2);$$

see [10, Proposition 5.5] and [11, Proposition 2.7]. We need two further assertions of that calculation. First, Heard identifies β with the image of the named stable β_1 under η [10, proof of Theorem 5.6]. Allowing for the choice of cohomology generator, this says

$$\eta_*(\beta_1) \in F^2 \pi_d Y, \quad [\eta_*(\beta_1)] = c\beta \in E_\infty^{2, 2ph}, \quad c \in \mathbb{F}_p^\times. \quad (2)$$

The brackets denote the class modulo F^3 . The Ext-to-group-cohomology comparison and permanence are also recorded in the discussion after [11, Proposition 2.7]. Second, in positive even filtration Heard’s ordinary E_∞ calculation is

$$E_\infty^{\text{even}, *} \cong \mathbb{F}_p[\Delta^{\pm p}][\beta] / (\beta^{h^2+1}), \quad \text{in positive filtration,} \quad (3)$$

by [10, Lemma 5.8]. In particular β^{h^2} is nonzero. These inputs hold for every odd prime; the named detection is not restricted to $p = 5$. They concern the extended group G and $\mathbb{F}_p$ coefficients. The pure subgroup and its larger coefficient field will be used separately in the exponent argument.

Lemma 3.2 (The ordinary filtered product). *Let a finite group Q act coherently on a commutative ring spectrum R . The ordinary homotopy fixed point spectral sequence has*

$$E_2^{s, t} = H^s(Q; \pi_t R)$$

and has the cup product on E_2 and the derivation rule for its differentials. Whenever this spectral sequence converges to $\pi_ R^{hQ}$ with a separated homotopy filtration whose successive quotients are E_∞ , its product on E_∞ is the associated-graded product for that actual homotopy filtration.*

The proof is given in Appendix A.

Proposition 3.3 (The actual product in the detector). *The unit $\eta : S \rightarrow Y$ carries $z_p = \beta_1^{h^2}$ to a nonzero class in $\pi_{dh^2} Y$.*

Proof. Apply Lemma 3.2 to the ordinary spectral sequence just described. The class of $\eta_*(\beta_1)^{h^2}$ in $F^{2h^2} \pi_{dh^2} Y / F^{2h^2+1} \pi_{dh^2} Y$ is

$$c^{h^2} \beta^{h^2} \neq 0$$

by (2) and (3). Hence the homotopy class itself is nonzero. Since η is a ring map, this power is exactly $\eta_*(\beta_1^{h^2})$, the image of the specified sphere product. $\square$

Proof of Theorem 3.1. The spectrum Y is $K(h)$ -local, since E_h is local and local objects are closed under homotopy limits. Its unit therefore factors through $L_{K(h)}S$, where the image of z_p must be nonzero by Proposition 3.3. Lemma 2.1 makes $L_{K(h)}(S \rightarrow X)$ an equivalence; naturality carries that nonzero image to $L_{K(h)}X$. For every $r \geq h$, the same lemma factors the detecting unit through $S \rightarrow X \rightarrow L_r X \rightarrow Y$. Its nonzero value on z_p proves the remaining assertion. $\square$

At $p = 5$ this is the named class β_1^{16} in degree $38 \cdot 16 = 608$. Section 5 kills its image at height five as part of the uniform upper argument.

4 Semilinear descent and an adic bound

At height $a = p$ for $p \geq 5$, and at height $a = p + 1$ for $p \geq 7$, the upper arguments need continuous cohomology to vanish above degree $a^2 + 1$, for both sphere coefficients and finite-test coefficients. We first bound the cohomology of finite adic quotients and then pass to their countable inverse limit. At height p the Galois quotient has order p , and its semilinear action on Witt coefficients supplies the required descent. At height $p + 1$ the full extended group has finite p -cohomological dimension.

Put $k_a = \mathbb{F}_{p^a}$ and write

$$(E_a)_* = O_a[u^{\pm 1}], \quad O_a = W(k_a)[[u_1, \dots, u_{a-1}]], \quad \mathfrak{m}_a = (p, u_1, \dots, u_{a-1}), \quad |u| = -2,$$

and $\mathbb{G}_a = \mathbb{S}_a \rtimes \text{Gal}(k_a/\mathbb{F}_p)$. Here $\mathbb{S}_a$ is the pure stabilizer over k_a .

Exactness of Galois invariants for p -complete twisted Witt modules is proved in [9, Lemma 1.14]. The explicit contraction below applies to arbitrary semilinear modules.

Lemma 4.1 (Trace-one contraction). *Let a finite group Q act on a commutative ring W . Suppose $w \in W$ satisfies $\sum_{g \in Q} g(w) = 1$. For any semilinear W -module P , one has $H^i(Q, P) = 0$ for $i > 0$, with no finiteness or flatness hypothesis on P . In particular this holds for $W = W(\mathbb{F}_{p^a})$ and $Q = \text{Gal}(\mathbb{F}_{p^a}/\mathbb{F}_p)$, without a prime-to- p hypothesis on a .*

Proof. Use homogeneous group cochains: a degree- q cochain is an equivariant function $f : Q^{q+1} \rightarrow P$, with differential the alternating deletion of coordinates. For $q \geq 1$ define

$$(hf)(g_0, \dots, g_{q-1}) = \sum_{\gamma \in Q} \gamma(w) f(\gamma, g_0, \dots, g_{q-1}).$$

For $g \in Q$, replacing γ by $g\gamma$ proves that hf is equivariant: semilinearity converts $(g\gamma)(w)gf$ into $g(\gamma(w)f)$. In $hd + dh$, the terms deleting one of the displayed g_i cancel in pairs. The term deleting the inserted coordinate is $\sum_{\gamma} \gamma(w)f = f$. Thus $hd + dh = \text{id}$ in positive degrees, proving acyclicity for arbitrary P .

For the Witt-ring case, the finite-field trace is onto. Lift a residue trace-one element to $w_0 \in W$. Its Witt trace lies in $\mathbb{Z}_p$ and is congruent to one modulo p , hence is a unit. Dividing w_0 by this invariant unit supplies w . We have not divided by $|Q|$. $\square$

Lemma 4.2 (Finite quotients and the extra adic degree). *Let $G = S_0 \rtimes Q$ be profinite with Q finite, and let $D \geq 0$ be an integer such that $\text{cd}_p(S_0) \leq D$ for discrete p -primary modules. Let $\{M_j\}_j$ be a countable tower of finite discrete p -primary G -modules with surjective transitions, and give $M = \varprojlim_j M_j$ the inverse-limit topology. Suppose each M_j is a W -module, S_0 acts W -linearly, Q acts semilinearly, and W admits the trace-one element of Lemma 4.1. Then*

$$H_{\text{cts}}^s(G, M_j) = 0 \quad (s > D), \quad H_{\text{cts}}^s(G, M) = 0 \quad (s > D + 1).$$

The second conclusion also holds if the first displayed finite-quotient vanishing is supplied directly, without a semidirect-product hypothesis.

Proof. The pure-subgroup cochain complex is W -linear and has semilinear Q -action, including the conjugation action on its arguments: for an inhomogeneous b -cochain,

$$(\gamma f)(s_1, \dots, s_b) = \gamma(f(\gamma^{-1}s_1\gamma, \dots, \gamma^{-1}s_b\gamma)).$$

The pure differential is W -linear and commutes with this action. Its cohomology is therefore semilinear, whether or not that cohomology has separately been shown finite. The continuous Hochschild–Serre sequence at the discrete coefficient level is

$$H^i(Q, H_{\text{cts}}^b(S_0, M_j)) \implies H_{\text{cts}}^{i+b}(G, M_j).$$

Lemma 4.1 kills its positive Q -columns, giving

$$H_{\text{cts}}^s(G, M_j) = H_{\text{cts}}^s(S_0, M_j)^Q.$$

This proves the first bound. Continuous cochains into M are compatible cochains into its finite quotients. Their transition maps are degreewise surjective: a continuous cochain has a finite value set, and any choice of lifts of those values gives a continuous cochain into the next finite quotient. No equivariant section is needed for inhomogeneous cochains. The countable inverse-limit sequence is therefore

$$0 \longrightarrow \varprojlim_j {}^1H_{\text{cts}}^{s-1}(G, M_j) \longrightarrow H_{\text{cts}}^s(G, M) \longrightarrow \varprojlim_j H_{\text{cts}}^s(G, M_j) \longrightarrow 0. \quad (4)$$

Both outer terms vanish for $s > D + 1$. At the remaining boundary degree, the same sequence gives

$$H_{\text{cts}}^{D+1}(G, M) \cong \varprojlim_j {}^1H_{\text{cts}}^D(G, M_j).$$

Thus the bound retains the possible extra inverse-limit degree. $\square$

Proposition 4.3 (The two upper heights). *Let V_0 be a finite ordinary p -local spectrum, let $DV_0 = F(V_0, S)$, and give $M = \pi_q(E_a \wedge DV_0)$ its $\mathfrak{m}_a$ -adic topology and Morava action. Then*

$$H_{\text{cts}}^s(\mathbb{G}_a, M) = 0 \quad (s > a^2 + 1)$$

in each of the following cases: $a = p$ with $p \geq 5$, and $a = p + 1$ with $p \geq 7$.

Proof. The pure stabilizer is the maximal-order unit group in the central division algebra over $\mathbb{Q}_p$ of dimension a^2 , so it is compact p -adic analytic of that dimension [24, Sections 2.1–2.2]. It has no element g of order p in either stated range. Indeed, $(g - 1)\Phi_p(g) = 0$ in a division algebra and $g \neq 1$ imply $\Phi_p(g) = 0$. The polynomial $\Phi_p(X + 1)$ is Eisenstein, so g generates the degree- $(p - 1)$ subfield $\mathbb{Q}_p(\zeta_p)$. The division algebra would then be a vector space over this subfield, forcing $p - 1 \mid a^2$. For $a = p$ this forces $p - 1 \mid 1$; for $a = p + 1$ it forces $p - 1 \mid 4$. These are impossible in the stated ranges. The Lazard–Serre theorem therefore gives $\text{cd}_p(\mathbb{S}_a) = a^2$ for discrete p -primary coefficients [29, Section 1, Corollary (1)].

For each q , the coefficient module M is finitely generated over the complete Noetherian ring O_a , by the finite construction of DV_0 . It is complete and separated; its quotients $M_j = M/\mathfrak{m}_a^j M$ are finite discrete p -primary modules with surjective transitions. This uses no flatness of M and allows torsion and both parities. The maximal ideal is invariant. For $V_0 = S$, continuity of the action on these quotients is part of the Morava action used by Devinatz–Hopkins [7, Theorem 1(ii) and Remark 1.3]. For every finite V_0 , the actual action and its continuity are also obtained by the ordinary cooperation construction of Proposition 8.1 below.

The pure stabilizer fixes $W(k_a)$. It fixes the residue field and thus the unique Hensel lifts of the roots of $X^{p^a} - X$, which are the Teichmüller representatives; continuity then fixes their convergent Witt expansions. Hence its action on M_j is Witt-linear, while $Q = \text{Gal}(k_a/\mathbb{F}_p)$ acts semilinearly by the Witt lift. At $a = p$, Lemma 4.2 with $D = p^2$ gives the bound. The full group has the split Galois subgroup of order p ; this conclusion is specific to the stated coefficients. The cohomological-dimension theorem was applied only to $\mathbb{S}_p$.

At $a = p + 1$ with $p \geq 7$, the Galois quotient has order $p + 1$, prime to p . An order- p element of $\mathbb{G}_a$ would therefore lie in $\mathbb{S}_a$, where we have just excluded it. The full group is a finite extension of the compact p -adic analytic pure group and has the same dimension a^2 . The same dimension theorem now applies to the full group itself and gives $H_{\text{cts}}^s(\mathbb{G}_a, M_j) = 0$ for $s > a^2$. Apply only the inverse-limit part of Lemma 4.2 to this finite-quotient bound. In both cases the finite quotients vanish for $s > a^2$, and (4) gives the asserted bound $s > a^2 + 1$ for M . $\square$

For the sphere-kernel argument, take $V_0 = S$, so that $M = (E_a)_q$. The exponent argument uses the same bound for every finite V_0 , including modules with torsion and both parities. In both applications the topology is the full maximal-ideal topology, as in the continuous-cochain convention of Devinatz–Hopkins [7, Remark 1.3].

Remark 4.4 (A sharper line for sphere coefficients). For the sphere homotopy fixed point spectral sequence, Bobkova and collaborators state an E_2 vanishing line at $s = n^2 + 1$ when $p - 1 \nmid n$ [5, Equation (2) and Remark 1.2]. At $p = n = 5$ this gives $H_{\text{cts}}^s(\mathbb{G}_5; (E_5)_q) = 0$ for $s > 25$. That statement concerns the ordinary sphere coefficients $(E_n)_q$. It does not assert the same line for every finite-test module $\pi_q(E_n \wedge DV_0)$ or identify an actual image filtration. Proposition 4.3 supplies the common conservative bound used in this paper.

5 Upper annihilation and the canonical kernels

At each upper height, torsion and parity place the unit image of β_1 in Adams filtration two. The cohomological bound of Section 4 makes the sufficiently high filtration zero. To apply this to the same power detected at the lower height, we identify the Adams filtration with images of tensor powers of the unit fiber. This also proves its multiplicativity directly.

5.1 Relative Adams stages

We first isolate the finite-stage comparison. In a stable symmetric monoidal ∞ -category with exact tensor, let E be a unital algebra. Call an object E -injective if it is a retract of $E \otimes Z$ for some Z . A relative E -Adams resolution of V consists of triangles

$$T_{s+1} \longrightarrow T_s \xrightarrow{q_s} Q_s, \quad T_0 = V,$$

where each Q_s is E -injective and $E \otimes q_s$ is split monic. All stages map to V by their composites down the tower. This is the relative convention of [7, Appendix A].

Lemma 5.1 (Equality of relative Adams-stage images). *Two such resolutions of V have the same stage- s image in $[W, V]$ for every object W and every $s \geq 0$.*

Proof. Suppose a map $f_s : T_s \rightarrow T'_s$ over V has been constructed, and put $g_s = q'_s f_s$. Choose a retract

$$Q'_s \xrightarrow{\iota_s} E \otimes Z_s \xrightarrow{\rho_s} Q'_s, \quad \rho_s \iota_s = 1,$$

and a splitting $\sigma_s : E \otimes Q_s \rightarrow E \otimes T_s$ with $\sigma_s(1_E \otimes q_s) = 1$. The composite

$$\begin{aligned} \bar{g}_s : Q_s &\xrightarrow{\eta_{Q_s}} E \otimes Q_s \xrightarrow{\sigma_s} E \otimes T_s \\ &\xrightarrow{1_E \otimes (\iota_s g_s)} E \otimes E \otimes Z_s \xrightarrow{\mu \otimes 1} E \otimes Z_s \xrightarrow{\rho_s} Q'_s \end{aligned} \quad (5)$$

extends g_s along q_s . Indeed, unit naturality gives $\eta_{Q_s} q_s = (1_E \otimes q_s) \eta_{T_s}$. The splitting removes $1_E \otimes q_s$, and a second use of unit naturality followed by the module unit law gives

$$\bar{g}_s q_s = \rho_s(\mu \otimes 1)(1_E \otimes (\iota_s g_s)) \eta_{T_s} = \rho_s \iota_s g_s = g_s.$$

Complete this square to a map of fiber triangles to obtain $f_{s+1} : T_{s+1} \rightarrow T'_{s+1}$. Starting with $f_0 = \text{id}_V$ gives maps over V at every finite stage, and hence one inclusion between the stage- s images in $[W, V]$. Repeating the construction in the other direction proves equality. The comparison begins at stage zero on V , so the indices agree. Only these finite-stage maps are used; no equivalence of towers is asserted. $\square$

5.2 The separated upper filtration

For a positive height a , work in the local category $\mathcal{C}_a = \text{Sp}_{K(a)}$, with

$$B_a = L_{K(a)} S, \quad U \otimes_a V = L_{K(a)}(U \wedge V), \quad I_a = \text{fib}(B_a \rightarrow E_a), \quad \epsilon_a : I_a \rightarrow B_a.$$

The tensor unit is B_a . These local tensors will be used to describe homotopy classes of the underlying ordinary spectra.

Lemma 5.2 (The actual Adams filtration). *The strongly convergent descent spectral sequence*

$$E_2^{s,q} = H_{\text{cts}}^s(\mathbb{G}_a; (E_a)_q) \implies \pi_{q-s} B_a \quad (6)$$

has the separated homotopy filtration

$$F^s \pi_* B_a = \text{im}(\pi_* I_a^{\otimes a s} \xrightarrow{(\epsilon_a^{\otimes a s})^*} \pi_* B_a), \quad I_a^{\otimes a 0} = B_a. \quad (7)$$

Its successive quotients are $E_\infty^{s,*}$ with the usual total degree, and $F^r F^s \subseteq F^{r+s}$.

Proof. Devinatz–Hopkins give (6) with strong convergence and identify it with the local E_a -Adams spectral sequence [7, Theorem 1(iii)–(iv)]; the full extended group is open in itself. The coefficient topology is the full maximal-ideal-adic topology of [7, Remark 1.3]. Their ideal $(p, v_1, \dots, v_{a-1})$ agrees degree-wise with $\mathfrak{m}_a = (p, u_1, \dots, u_{a-1})$, since $v_i = u_i u^{1-p^i}$ and u is invertible.

The tensor powers $T_s = I_a^{\otimes a s}$ form resolution triangles

$$T_{s+1} \longrightarrow T_s \xrightarrow{q_s} E_a \otimes_a T_s. \quad (8)$$

Their right terms are E_a -injective, and multiplication on E_a retracts $E_a \otimes_a q_s$. Thus they form a relative Adams resolution in the convention just used. By Lemma 5.1, their stage- s images agree with those in the Devinatz–Hopkins resolution at the same index. This proves (7). In particular $T_0 = B_a$ and $T_1 = I_a$ give the full group and the kernel of $\pi_* B_a \rightarrow \pi_* E_a$, respectively. Strong convergence supplies separation and the stated associated-graded quotients for these actual image groups.

A map from a suspended sphere to a local object is equivalently a map from the suspended local unit. Tensoring two lifts through $I_a^{\otimes a r}$ and $I_a^{\otimes a s}$ therefore gives a lift of their product through $I_a^{\otimes a (r+s)}$. This proves multiplicativity. Monoidal localization carries the sphere unit and its products to the local unit, so these are also the products of the actual sphere images. Smashing localization is not needed. $\square$

Proposition 5.3 (The zero upper tail). *Let $a = p$ with $p \geq 5$, or $a = p + 1$ with $p \geq 7$. Then*

$$F^s \pi_* B_a = 0 \quad (s \geq a^2 + 2). \quad (9)$$

Proof. Proposition 4.3, with the finite test $V_0 = S$, gives

$$H_{\text{cts}}^s(\mathbb{G}_a; (E_a)_q) = 0 \quad (s > a^2 + 1, q \in \mathbb{Z}). \quad (10)$$

At height p this uses the pure stabilizer dimension and the semilinear Witt trace-one contraction for the order- p Galois quotient. At height $p + 1$ the full extended group has no p -torsion, so its dimension gives the finite-quotient bound. Both cases retain the possible adic inverse-limit contribution in degree $a^2 + 1$.

By Lemma 5.2, the successive quotients of the actual filtration from $a^2 + 2$ onward are zero. That tail is consequently constant, and separatedness makes its common value zero. This proves (9). It is a statement about homotopy images; the exponent argument will require a separate proof of map nullity. $\square$

5.3 Annihilating the detected sphere product

Put $d = 2p(p - 1) - 2$ and $k = (p - 1)^2$.

Proposition 5.4 (Upper vanishing of the same power). *For every $x \in \pi_d S$, the unit image of x^k is zero in $\pi_{dk} B_p$ when $p \geq 5$. When $p \geq 7$, the image of the same product is also zero in $\pi_{dk} B_{p+1}$.*

Proof. Let a be either upper height in its stated range, and let $x_a \in \pi_d B_a$ be the unit image of x . Positive stable sphere stems are torsion: suspend into an odd-dimensional sphere in the stable range and apply Serre's finiteness theorem [28, Chapter V, Section 3, Proposition 3]. Since $(E_a)_d$ is torsion-free, x_a maps to zero in $\pi_d E_a$ and belongs to $F^1 \pi_d B_a$.

Since d is even and E_a has only even homotopy,

$$F^1 \pi_d B_a / F^2 \pi_d B_a \cong E_\infty^{1, d+1} = 0.$$

Thus $x_a \in F^2$, and multiplicativity in Lemma 5.2 gives $x_a^k \in F^{2k} \pi_{dk} B_a$. The numerical margins above the last possible filtration are

$$\begin{aligned} 2(p - 1)^2 - (p^2 + 1) &= p^2 - 4p + 1 > 0 \quad (p \geq 5), \\ 2(p - 1)^2 - ((p + 1)^2 + 1) &= p(p - 6) > 0 \quad (p \geq 7). \end{aligned}$$

Therefore $2k \geq a^2 + 2$, and (9) gives $x_a^k = 0$. This is the unit image of the original sphere product x^k , by the product identification in Lemma 5.2. $\square$

Proof of Theorems 1.2 and 1.1, and Corollary 1.3. Use the single named product $z_p = \beta_1^{(p-1)^2}$. Proposition 3.3 detects it under the actual unit to the $K(p - 1)$ -local spectrum $Y = E_{p-1}^{hG}$. Proposition 5.4, applied to $x = \beta_1$, kills that same product in B_p for $p \geq 5$ and in B_{p+1} for $p \geq 7$.

Apply Proposition 2.2 with $t = p - 1$ and $n = p$ or $n = p + 1$, respectively. The result is a nonzero image of z_p in $\pi_{dk} L_{n-1} X$ killed by the actual canonical map $i_{n, X}$. Its two naturality squares identify that map after completion. A retraction would induce a left inverse on this homotopy group, which is impossible.

For $p = 5$ the product is β_1^{16} in degree 608, and its upper image lies in $F^{32} \pi_{608} B_5$, while the zero tail starts at F^{27} . Taking $n = 5$ gives the named prime-five statement as well. $\square$

6 Finite descent and exponent bounds

The exponent argument uses two consequences of a finite descent bound. First, such a bound forces a horizontal vanishing line on a specified page of the descent spectral sequence. Second, an E_2 vanishing line, together with some finite descent bound, forces a specified fiber map to have zero image after every chosen exact test. We prove these statements directly from finite tensor cubes, keeping tested images distinct from null maps.

6.1 Finite tensor cubes

Let $\mathcal{C}$ be a stable symmetric monoidal ∞ -category whose tensor is exact in each variable. Let A be an associative unital algebra, with unit $\mathbf{1}$, and put $J = \text{fib}(\mathbf{1} \rightarrow A)$ and $\epsilon : J \rightarrow \mathbf{1}$. For $V \in \mathcal{C}$ form the augmented Amitsur object

$$V \longrightarrow C_A^\bullet(V), \quad C_A^q(V) = A^{\otimes(q+1)} \otimes V.$$

Cofaces insert the unit and codegeneracies multiply adjacent A -factors. The partial totalization Tot_s is the limit over the full simplex category on $[0], \dots, [s]$, including codegeneracies. The cubical comparison is standard descent machinery [21, Proposition 2.14, pp. 1007–1008]; its proof fixes the finite-stage indexing and functoriality used here.

Lemma 6.1 (Full finite tensor cube). *For $s \geq 0$ there is a natural equivalence of arrows over id_V in $\mathcal{C}_V$:*

$$\begin{aligned} (\text{fib}(V \rightarrow \text{Tot}_s C_A^\bullet(V)) &\longrightarrow V) \\ &\simeq \left(J^{\otimes(s+1)} \otimes V \xrightarrow{\epsilon^{\otimes(s+1)} \otimes \text{id}_V} V \right). \end{aligned}$$

The transition on the fibers from stage $s+1$ to stage s applies ϵ to the final J -factor. Any exact functor $\Phi : \mathcal{C} \rightarrow \text{Sp}$ preserves this finite fiber sequence.

Proof. Let $\mathcal{P}_{\neq \emptyset}([s])$ be the poset of nonempty subsets of $\{0, \dots, s\}$. The functor

$$\theta_s : \mathcal{P}_{\neq \emptyset}([s]) \longrightarrow \Delta_{\leq s}, \quad S \longmapsto [|S| - 1]$$

uses the order-preserving injection induced by each inclusion of subsets. At $[l]$, with $l \leq s$, the comma category $\theta_s/[l]$ consists of nonempty subsets equipped with a weakly increasing map to $[l]$. It is the poset of nonempty faces of the simplicial complex $P(s, l)$ whose simplices are lists

$$(i_0, j_0), \dots, (i_b, j_b), \quad 0 \leq i_0 < \dots < i_b \leq s, \quad 0 \leq j_0 \leq \dots \leq j_b \leq l.$$

This complex is contractible. The base case $P(0, 0)$ is a point. Every simplex either omits the first-coordinate value s or contains exactly one vertex (s, j) . Begin inductively with the cone of vertex (s, l) on $P(s-1, l)$. This cone is contractible even when $l = s$, when its base is not covered by induction. For each $j < l$, attach the cone of vertex (s, j) along its base $P(s-1, j)$. This is its exact intersection with the previous complex and is contractible by induction, since $j \leq s-1$. The intersections are simplicial subcomplexes, so each attachment preserves contractibility. The functor is therefore homotopy initial.

The partial totalization is consequently the limit of the punctured $(s+1)$ -cube obtained by tensoring copies of $\mathbf{1} \rightarrow A$ and then tensoring with V . Its omitted vertex is V . Taking successive fibers and using exactness of tensor gives $J^{\otimes(s+1)} \otimes V$, with its canonical arrow to V . Removing the last direction of the cube gives the stated transition. This is a genuinely finite cubical limit, so any exact Φ preserves it. Finiteness of the list of objects in $\Delta_{\leq s}$ alone would not justify that assertion. $\square$

Fix an exact functor $\Phi : \mathcal{C} \rightarrow \mathbf{Sp}$. Put $V' = \Phi V$, $U_l = \Phi(J^{\otimes l} \otimes V)$, and $P_s = \text{Tot}_s(\Phi C_A^\bullet(V))$ for $s \geq 0$; set $P_s = 0$ for $s < 0$. Thus $U_0 = V'$ and

$$U_{s+1} \longrightarrow V' \longrightarrow P_s \quad (11)$$

is a compatible family of fiber sequences. Define its augmentation filtration by

$$F^s \pi_d V' = \ker(\pi_d V' \rightarrow \pi_d P_{s-1}) = \text{im}(\pi_d U_s \rightarrow \pi_d V'), \quad s \geq 0. \quad (12)$$

These groups are defined from finite stages. The argument below will use them without identifying V' with an infinite totalization. Our spectral sequence convention is

$$E_2^{s,q} = H^s \pi_q(\Phi C_A^\bullet(V)), \quad d_r : (s, q) \mapsto (s+r, q+r-1).$$

The full tower has normalized cosimplicial cochains on E_1 , giving the displayed E_2 page.

6.2 Exponents and exact finite-page bounds

For $b \geq 1$ say that V has A -exponent at most b if the actual map

$$e_b(V) = \epsilon^{\otimes b} \otimes \text{id}_V : J^{\otimes b} \otimes V \longrightarrow V$$

is null. This is stronger than vanishing on homotopy groups after a test.

Lemma 6.2 (Exponent calculus). *An A -module has exponent at most one. Bounds are preserved by retracts and by tensoring with arbitrary objects. For a cofiber sequence $U \rightarrow V \rightarrow W$, bounds b and c on U and W give bound $b+c$ on V . If $e_m(V)$ is null, V is a retract of an object with m successive layers $A \otimes J^{\otimes i} \otimes V$, $0 \leq i < m$.*

Proof. The unit map of a module splits by its action, making its fiber map null. Naturality gives retract invariance, and tensoring a null map gives tensor invariance. In the cofiber sequence, the composite $J^{\otimes c} \otimes V \rightarrow V \rightarrow W$ is null, so choose a lift $\ell : J^{\otimes c} \otimes V \rightarrow U$. After precomposition by b more ϵ factors, naturality identifies the resulting map into U with $e_b(U)(1 \otimes \ell)$, which is null. Its composite to V is $e_{b+c}(V)$. Finally the consecutive maps $J^{\otimes(i+1)} \otimes V \rightarrow J^{\otimes i} \otimes V$ have the stated cofibers. Their composites to V filter $\text{cofib}(e_m(V))$ by these m layers. Nullity of $e_m(V)$ makes V a retract of that cofiber. $\square$

These bounds describe the thick tensor ideal generated by A ; compare [20, Definition 3.18]. We now fix the page convention for the quantitative use of nilpotence [21, 22].

Proposition 6.3 (Pages and tested images). *For any exact $\Phi : \mathcal{C} \rightarrow \mathbf{Sp}$ and integer $b \geq 1$:*

- (i) *If V has A -exponent at most b , then $E_{b+1}^{s,q} = 0$ for all $s \geq b$ and q .*
- (ii) *If V has some finite A -exponent and $E_2^{s,q} = 0$ for all $s \geq b$ and q , then $\pi_d \Phi(e_b(V)) = 0$ for every d .*

The second assertion is zero tested image, not nullity of a map.

Proof. In coordinates (s, d) with $d = q - s$, the derived couple has

$$D_r^{s,d} = \text{im}(\pi_d P_{s+r-1} \rightarrow \pi_d P_s)$$

and maps

$$i : D_r^{s,d} \rightarrow D_r^{s-1,d}, \quad j : D_r^{s,d} \rightarrow E_r^{s+r,d-1}, \quad k : E_r^{s,d} \rightarrow D_r^{s,d}.$$

Here the second E -index denotes total degree; no convergence claim is used. Let c be an actual exponent bound. Every length- c transition in the fiber tower U_l is null, being Φ applied to a tensor multiple of $e_c(V)$. For $s \geq 0$ the boundary square from (11) is

$$\begin{array}{ccc} \pi_d P_{s+c} & \longrightarrow & \pi_{d-1} U_{s+c+1} \\ \downarrow & & \downarrow 0 \\ \pi_d P_s & \longrightarrow & \pi_{d-1} U_{s+1}. \end{array}$$

The image from P_{s+c} has zero boundary and lifts to $\pi_d V'$; conversely an augmentation class comes from P_{s+c} . Hence

$$D_{c+1}^{s,d} = \text{im}(\pi_d V' \rightarrow \pi_d P_s). \quad (13)$$

Both sides are zero for negative s . These image groups have surjective transition i . The augmentation into P_s is injective on homotopy for $s \geq c-1$, since its kernel is the image of a map factoring through $e_c(V)$. Thus i is an isomorphism for $s \geq c$. Exactness of the derived couple gives $j = 0$ and, for $s \geq 0$, identifies

$$E_{c+1}^{s,d} \cong \ker(D_{c+1}^{s,d} \rightarrow D_{c+1}^{s-1,d}) \cong F^s \pi_d V' / F^{s+1} \pi_d V'.$$

In particular this is zero for $s \geq c$. Taking $c = b$ proves (i). For (ii) retain any c . The E_2 line makes these same quotients zero for $s \geq b$. Since $F^c = 0$, descending from c , or monotonicity when $c < b$, gives $F^b = 0$. Equation (12) is exactly the asserted zero image. This argument uses only finite cubes and never assumes that Φ preserves an infinite limit. $\square$

7 A finite-page lower exponent bound

Fix a prime $p \geq 5$ and put $t = p-1$, $h = p$, and $m = p^2 + 2$. The lower exponent inequality in Theorem 1.4 will follow from a class in filtration greater than m on page E_{m+1} . Proposition 6.3(i) excludes such a class whenever the exponent is at most m . This argument uses the pure stabilizer subgroup and its residue field, which we specify independently of the extended group used for the sphere kernel.

Theorem 7.1 (Pure-stabilizer cohomology). *For every odd prime p , put $t = p-1$ and let E_t be Morava E -theory for the height- t Honda formal group over $k_t = \mathbb{F}_{p^t}$, with its coherent stabilizer action. The pure stabilizer $\mathbb{S}_t$ contains a finite subgroup $H \cong C_p \rtimes C_{t^2}$ for which*

$$\begin{aligned} \hat{H}^*(H; (E_t)_*) &= k_t[\alpha, \beta^{\pm 1}, \Delta^{\pm 1}] / (\alpha^2), \\ |\alpha| &= (1, 2t), \quad |\beta| = (2, 2pt), \quad |\Delta| = (0, 2pt^2). \end{aligned} \quad (14)$$

The ordinary-to-Tate comparison is multiplicative and identifies $H^s(H; (E_t)_q)$ with the displayed Tate group for every $s > 0$ and q .

The coherent action specializes to the prescribed Honda stabilizer action [7, Theorem 1(ii) and Remark 2.4]. This is the subgroup denoted F in [1, Equation (1.2) and Theorem 4.5], where the height is called n . The positive-filtration comparison is [1, Remark 2.2]; the pure-group calculation before taking Galois invariants also appears in [10, proof of Proposition 5.5]. We use the cohomology and its bidegrees here. No differential from the Tate spectral sequence is needed for the argument below.

Work in $\mathcal{D}_H = \text{Fun}(BH, \text{Sp}_{(p)})$ with pointwise ordinary smash, and put $A_H = F(H_+, S)$ with its translation action. For the exact functor $(-)^{hH}$, the A_H -Amitsur spectral sequence is the ordinary homotopy fixed point spectral sequence: its terms are functions on the free H -sets H^{q+1} , and taking fixed points gives homogeneous group cochains. Lemma 3.2 supplies its cup product and derivation rule without requiring convergence. We will apply the finite-page exponent criterion to this tower.

Proposition 7.2. *The H -exponent of E_t is greater than m .*

Proof. We have $m < 2t^2 + 1$, since the difference is $p^2 - 4p + 1 > 0$. Choose a positive N divisible by p with $2N > m$. We show that $\beta^N \neq 0$ on E_{m+1} in bidegree $(2N, 2ptN)$. For $2 \leq r \leq m$ all possible source and target filtrations are positive, so (14) applies. Even r gives odd internal degree and a zero group. For odd r , internal-degree divisibility forces $r = 2ta + 1$ with $1 \leq a < t$.

An outgoing target must be $\alpha\beta^{N+ta}\Delta^l$. Equality of internal degrees gives $a - 1 = pt(a + l)$, which forces $a = 1$. An incoming source must be $\alpha\beta^{N-ta-1}\Delta^l$, whose internal-degree equation is $a - t = pt(a - l)$, impossible for $1 \leq a < t$. Thus only the outgoing differential d_{2t+1} remains. The same degree argument shows that β exists up to that page; its only possible incoming differential has length two and odd internal source degree. The derivation rule on that page gives

$$d_{2t+1}(\beta^N) = N\beta^{N-1}d_{2t+1}(\beta) = 0,$$

because β has even total degree, p divides N , and the target is p -torsion. With all incoming differentials and all other outgoing possibilities excluded through page m , this class is nonzero on E_{m+1} . Its filtration is $2N > m$, contradicting Proposition 6.3(i) if the exponent were at most m . $\square$

At $p = 5$, where $m = 27$, one may take $N = 15$: the class has filtration 30 on E_{28} and obstructs exponent 27. The proof needs only this finite-page survival; it does not use a sphere representative or the abutment of the ordinary spectral sequence.

8 Ordinary cooperations and finite tests

The upper exponent bound requires the cohomological line of Section 4 after every finite ordinary test. We therefore identify the tested Amitsur coefficient complex, including its actual coface and codegeneracy maps. Fix a positive height a , and write

$$\begin{aligned} E &= E_a, & k_a &= \mathbb{F}_{p^a}, & O &= O_a = W(k_a)[[u_1, \dots, u_{a-1}]], \\ O_* &= (E_a)_* = O[u^{\pm 1}], & |u| &= -2, & \mathfrak{m} &= \mathfrak{m}_a = (p, u_1, \dots, u_{a-1}). \end{aligned}$$

Use the coherent action of $\mathbb{G}_a = \mathbb{S}_a \rtimes \text{Gal}(k_a/\mathbb{F}_p)$ by commutative ring automorphisms, whose special-fiber action is the prescribed Honda stabilizer action [8, Section 7, Proposition 7.3, Equation (7.3), and Corollaries 7.6–7.7]. The local category $\mathcal{C}_a = \text{Sp}_{K(a)}$ has unit $B_a = L_{K(a)}S$ and tensor $U \otimes_a V = L_{K(a)}(U \wedge V)$. All unadorned smash products and function spectra below are ordinary.

Proposition 8.1 (The finite-test coefficient complex). *For $r \geq 0$, let*

$$Z_r = L_{K(a)}(E^{\wedge(r+1)}).$$

For every finite ordinary p -local spectrum W , there are natural graded isomorphisms

$$\pi_*(Z_r \wedge W) \cong C_{\text{cts}}(\mathbb{G}_a^r, \pi_*(E \wedge W)). \quad (15)$$

The first E -factor acts by constant coefficients. The two-factor evaluation is multiplication after $1 \wedge g$. In the cumulative coordinates $1, g_1, g_1 g_2, \dots, g_1 \cdots g_r$ on the factors, these isomorphisms identify every coface and codegeneracy of the unit Amitsur object with the continuous inhomogeneous group-cochain operator.

Consequently, apply the exact ordinary functor $F(V_0, -)$, for a finite ordinary p -local spectrum V_0 , to the full unit Amitsur tower for E_a in $\mathcal{C}_a$. In the page convention of Proposition 6.3, its second page is

$$E_2^{s,q} = H_{\text{cts}}^s(\mathbb{G}_a, \pi_q(E_a \wedge DV_0)), \quad DV_0 = F(V_0, S). \quad (16)$$

Here H_{cts}^s means the cohomology of continuous cochains with the adic topology on the indicated coefficient module.

We prove the comparison by first forming ordinary cooperations, then completing their flat coefficient modules, and finally applying the finite test. The next three lemmas justify these steps. Mixed cooperations and their completion will also be used in Section 10. The exact-homology input originates in Landweber's work [17]; we use the precise cooperation formulations cited below, with BP the p -local Brown–Peterson spectrum.

Lemma 8.2 (Ordinary mixed cooperations). *For positive heights a, b , the ordinary smash product has even homotopy*

$$\pi_*(E_a \wedge E_b) \cong (E_a)_* \otimes_{BP_*} BP_* BP \otimes_{BP_*} (E_b)_*.$$

It is flat over $(E_a)_$ through the first factor.*

Proof. Apply the ordinary Landweber cooperation calculation to the p -typical flat Hopf algebroid $(BP_*, BP_* BP)$. This gives the displayed BP_* formula. It is the p -typical form of the MU_* formulas displayed in [26, Section 15, Proposition 15.1 and Corollary 15.4] and [14, Lemma 4.4]; Morava E -theory over BP is included in [16, Example 0.1(d)]. The two tensor products use the two different unit maps into $BP_* BP$. For a Landweber-exact BP_* -algebra C , the module $BP_* BP \otimes_{BP_*} C$ is flat over the other BP_* -action. Indeed, tensoring with it factors as the exact cofree-comodule functor followed by the exact Landweber functor on comodules. This is also the opposite-unit criterion of [16, Lemma 2.2], with conjugation exchanging the units when necessary. Base change along $BP_* \rightarrow (E_a)_*$ proves the asserted flatness. The displayed coefficient and cooperation rings are even graded, which proves evenness. In particular this argument does not require $(E_b)_*$ to be flat over BP_* . $\square$

The ring O is complete, Noetherian, regular, and local, with finite residue field. If W is a finite ordinary p -local spectrum, then

$$M_* = \pi_*(E \wedge W)$$

is finitely generated over O_* . This follows by finite cofiber constructions and retracts from the assertion for the sphere. Periodicity identifies graded module questions with the two parities over O . Regularity therefore supplies a finite resolution [30, Tag 00O7] of M_* by finite graded free O_* -modules. Neither parity of M_* is required to vanish.

Lemma 8.3 (Base change and continuous functions). *Let D_0 be an ordinary right E -module for which $\pi_* D_0 \otimes_{O_*} (-)$ is exact on finitely generated graded O_* -modules. The natural comparison*

$$\pi_* D_0 \otimes_{O_*} \pi_*(E \wedge W) \longrightarrow \pi_*(D_0 \wedge W) \quad (17)$$

is an isomorphism for every ordinary p -local spectrum W .

For every profinite set T , the functor $C_{\text{cts}}(T, -)$ is exact on finitely generated O -modules with their $\mathfrak{m}$ -adic topologies. For every finitely generated graded O_* -module M_* , the natural map is an isomorphism

$$C_{\text{cts}}(T, O_*) \otimes_{O_*} M_* \xrightarrow{\cong} C_{\text{cts}}(T, M_*). \quad (18)$$

The functions and the displayed graded isomorphism are understood degreewise.

Proof. First suppose W finite. The equality

$$D_0 \wedge W \simeq D_0 \wedge_E (E \wedge W)$$

is an equality of ordinary derived constructions. The comparison is the balanced homotopy cross product for this relative smash, so it is natural in both variables before choosing any resolution. Realize a finite graded free surjection onto $\pi_*(E \wedge W)$ by an E -module map from a finite wedge of suspensions of E . Surjectivity makes the homotopy of its fiber the first syzygy. Repeat along a finite free resolution. At the last stage, lifts of a free basis give an equivalence from a finite free E -module, since they give an isomorphism on all homotopy groups. Relative smash with D_0 is exact. Induction back through these fiber sequences, using the assumed tensor exactness on each finitely generated syzygy, proves (17) for finite W .

Every ordinary p -local spectrum is a filtered homotopy colimit of finite ordinary p -local spectra. Both sides of (17) preserve those colimits: ordinary smash and homotopy groups do so, as does tensoring a module with $\pi_* D_0$. Naturality extends the comparison to arbitrary W .

For the assertion about functions, let $M \rightarrow N$ be a surjection of finitely generated O -modules. For every $j \geq 1$ the map

$$M/\mathfrak{m}^{j+1}M \longrightarrow (M/\mathfrak{m}^jM) \times_{N/\mathfrak{m}^jN} (N/\mathfrak{m}^{j+1}N)$$

is surjective. To see this, lift the first component to M ; its error against the second component lies in $\mathfrak{m}^jN$, and this error lifts from $\mathfrak{m}^jM$. Each quotient is a finite set. Given a continuous map $T \rightarrow N$, first lift its reduction modulo $\mathfrak{m}$, and then use these surjections to lift successively while preserving the previous choice. A lift on each finite value set is continuous. The compatible system defines a continuous lift to M , since finitely generated modules are complete. Kernels carry their subspace topologies: the relevant modules are compact Hausdorff, and the induced continuous bijection from a kernel to its image is a homeomorphism. It follows that $C_{\text{cts}}(T, -)$ is exact on these modules. Apply this exact functor to a finite presentation of M , and compare with tensoring the same presentation by $C_{\text{cts}}(T, O)$. The comparison is an isomorphism on the finite free terms and hence on M . Applying this argument in each parity proves (18). In particular $C_{\text{cts}}(T, O_*) \otimes_{O_*} (-)$ has precisely the tensor exactness used in the first assertion. $\square$

Lemma 8.4 (Flat completion and reduction). *Let N be an ordinary E -module with flat graded homotopy $U_* = \pi_* N$. Its actual $K(a)$ -localization map induces*

$$U_* \longrightarrow \pi_* L_{K(a)} N \cong \widehat{U}_* := \varprojlim_j U_*/\mathfrak{m}^j U_*.$$

For a flat O -module U and every $j \geq 1$,

$$\widehat{U}/\mathfrak{m}^j \widehat{U} \cong U/\mathfrak{m}^j U.$$

Both assertions apply degreewise to the periodic graded modules above.

Proof. The first assertion is the localization theorem for ordinary Morava E -modules: $K(a)$ -localization is derived $\mathfrak{m}$ -completion, and flat homotopy has no higher derived completion. The natural localization map induces the ordinary completion map [27, Propositions 3.4, 3.6 and Corollary 3.8].

Here is the algebraic reduction argument, including its needed inverse-limit step. Choose a resolution $P_\bullet \rightarrow N_0$ of a finitely generated O -module N_0 by finite free modules. The tower $P_\bullet \otimes_O U/\mathfrak{m}^l U$ is degreewise surjective. By flatness of U , its first homology is

$$\mathrm{Tor}_1^O(N_0, O/\mathfrak{m}^l) \otimes_O U.$$

This homology tower is pro-zero. Indeed, for a finite first presentation $0 \rightarrow K \rightarrow P_0 \rightarrow N_0 \rightarrow 0$, the corresponding Tor group is $(K \cap \mathfrak{m}^l P_0)/\mathfrak{m}^l K$. Artin–Rees [30, Tag 00IN] gives a d such that $K \cap \mathfrak{m}^l P_0 \subseteq \mathfrak{m}^{l-d} K$ for $l \geq d$. For any fixed j , a transition from $l \geq j + d$ to j is therefore zero. Tensoring these zero transitions with U preserves this conclusion.

The homology exact sequence for the countable inverse limit of this degreewise-surjective tower now gives

$$N_0 \otimes_O \widehat{U} \cong \varprojlim_l (N_0 \otimes_O U/\mathfrak{m}^l U);$$

the possible $\varprojlim^1$ of the first homology is zero because that tower is pro-zero. Finite freeness of the P_i identifies their tensor with $\widehat{U}$ with their inverse-limit tensor. Take $N_0 = O/\mathfrak{m}^j$; for $l \geq j$ the modules on the right are $U/\mathfrak{m}^j U$. This proves the reduction formula. $\square$

Proof of Proposition 8.1. The completed cooperation theorem gives

$$\pi_* L_{K(a)}(E \wedge E) \cong C_{\mathrm{cts}}(\mathbb{G}_a, O_*). \quad (19)$$

At g , the map is $\mu(1 \wedge g)$; the first factor acts by constant scalars. This is the evaluation map of [14, Section 1 and Theorems 4.11 and 6.8]. The equivalent convention in [7, Proposition 2.2, Equation (2.3), and Equations (2.5)–(2.7)] uses $\mu(g^{-1} \wedge 1)$; exchanging the factors and inverting the group coordinate gives the convention above. The coefficient topology is the adic one [7, Remark 1.3].

Put $P_* = \pi_*(E \wedge E)$, regarded as an O_* -module through the first factor. Lemma 8.2 makes P_* flat. The actual localization map, the reduction in Lemma 8.4, and (18) applied to $O_*/\mathfrak{m}^j O_*$ give

$$P_*/\mathfrak{m}^j P_* \cong C_{\mathrm{cts}}(\mathbb{G}_a, O_*/\mathfrak{m}^j O_*). \quad (20)$$

This isomorphism is induced by the actual ordinary evaluation $E \wedge E \xrightarrow{1 \wedge g} E \wedge E \xrightarrow{\mu} E$ followed by reduction. It is not just an abstract isomorphism of the two reduction modules.

To form higher powers, use a common first factor in the ordinary relative smash products

$$(E_0 \wedge E_1) \wedge_{E_0} \cdots \wedge_{E_0} (E_0 \wedge E_r) \simeq E_0 \wedge E_1 \wedge \cdots \wedge E_r.$$

Each E_i denotes the same spectrum E ; the subscripts mark positions. Iterating (17), including its extension to arbitrary ordinary W , identifies the homotopy of this ordinary smash with $P_*^{\otimes_{O_*} r}$. The comparison inserts each pair into the first and one other position and multiplies at the common position. These modules are flat, since P_* is flat.

In each degree the coefficient ring $O_*/\mathfrak{m}^j O_*$ is finite. Continuous functions to it are unions of function modules on finite quotient sets of $\mathbb{G}_a$. Tensors of the function modules on finite sets are the function modules on their products, and every continuous function on $\mathbb{G}_a^r$ to a finite set factors

through finite quotients in the coordinates. It follows from (20) that the reductions of $P_*^{\otimes_{O_*} r}$ are $C_{\text{cts}}(\mathbb{G}_a^r, O_*/\mathfrak{m}^j O_*)$. Apply Lemma 8.4 only now, to this ordinary $(r+1)$ -fold smash. The result is

$$\pi_* Z_r \cong C_{\text{cts}}(\mathbb{G}_a^r, O_*). \quad (21)$$

For $r = 0$ this is the locality of E .

The maps in (21) remain the evaluations on the non-first factors followed by multiplication. In fact each ordinary evaluation extends through localization because its target E is local. On homotopy its first-factor O_* -linearity makes it adically continuous. Lemma 8.4 makes the ordinary image dense, and the reductions above show that the extension agrees with the displayed continuous function. Thus the construction preserves the evaluation maps as well as the coefficient modules.

For finite W , the ordinary E -module Z_r satisfies the exactness assumption in Lemma 8.3, by (21) and (18). Those comparisons give (15). This step allows any finitely generated $\pi_*(E \wedge W)$, including torsion and both parities; no flatness of that module has been used.

The terms of the local unit Amitsur object are exactly Z_r . For finite V_0 , ordinary duality identifies $F(V_0, Z_r) \simeq Z_r \wedge DV_0$, so the previous formula computes each cosimplicial homotopy group. Relabel the independent evaluations by their cumulative coordinates

$$1, \quad g_1, \quad g_1 g_2, \quad \dots, \quad g_1 \cdots g_r.$$

The cochain identities below can be checked on ordinary evaluations before localization. For finite W , their target $E \wedge W$ is $K(a)$ -local, since finite dualizable smash preserves local objects. They therefore extend through localization and remain identities on the tested coefficients. Insertion of the first unit acts on the coefficient; an interior unit combines two consecutive coordinates, and insertion of the last unit drops the last coordinate. Explicitly, for a degree- r cochain c ,

$$\begin{aligned} (\delta^0 c)(g_1, \dots, g_{r+1}) &= g_1 c(g_2, \dots, g_{r+1}), \\ (\delta^i c)(g_1, \dots, g_{r+1}) &= c(g_1, \dots, g_i g_{i+1}, \dots, g_{r+1}) \quad (1 \leq i \leq r), \\ (\delta^{r+1} c)(g_1, \dots, g_{r+1}) &= c(g_1, \dots, g_r). \end{aligned}$$

Multiplication of adjacent factors gives each codegeneracy, which inserts the identity in the corresponding coordinate. These are all the inhomogeneous cochain operators, including the degeneracies of the full cosimplicial object. The first unit insertion also displays the orbit function of every coefficient. Those functions are continuous, and reduction gives continuous actions on the finite invariant quotients by powers of $\mathfrak{m}_a$. The E_2 -page of the tower of finite partial totalizations is cosimplicial cohomology of these homotopy groups, which is (16). $\square$

9 From finite tests to an actual null map

We now bound the exponent of the upper local unit by proving nullity of a specified tensor-power map.

Proposition 9.1 (The upper null map and its finite layers). *Let $p \geq 5$ be prime and put $h = p$. In $\mathcal{C}_h$, let $I_h = \text{fib}(B_h \rightarrow E_h)$. The actual map*

$$e_m(B_h) : I_h^{\otimes_h m} \longrightarrow B_h, \quad m = h^2 + 2 = p^2 + 2,$$

is null. Thus $\exp_{E_h}(B_h) \leq m$, and B_h is a retract of an object with m successive layers

$$E_h \otimes_h I_h^{\otimes_h i}, \quad 0 \leq i < m.$$

The coefficient line will make this map zero after every finite ordinary precomposition, once some finite exponent is known. The following property of its target then turns that tested vanishing into nullity. Its relation to the Brown–Comenetz phantom criterion is [6, Proposition 4.11].

Lemma 9.2 (Ordinary phantoms into a local sphere). *Let $a \geq 1$. An ordinary map $f : U \rightarrow B_a$ of p -local spectra is null if its precomposition with every map from a finite ordinary p -local spectrum to U is null.*

Proof. Let $M_a S = \text{fib}(L_a S \rightarrow L_{a-1} S)$, and let I_{BC} be the ordinary Brown–Comenetz dualizing spectrum for the injective group $\mathbb{Q}/\mathbb{Z}_{(p)}$. Gross–Hopkins duality, announced in [12, Theorem 6] and proved in the form used here in [15, Theorem 10.2(b),(e)], says that

$$D_a = F(M_a S, I_{\text{BC}})$$

is $K(a)$ -local and invertible for $\otimes_a$. The identification with this monochromatic object uses [15, Theorem 6.19].

For a local target Z , the ordinary spectrum $F(R, Z)$ is local. A map into it from a $K(a)$ -acyclic spectrum V corresponds to a map $V \wedge R \rightarrow Z$, and $V \wedge R$ is still acyclic. Hence ordinary function spectra compute the local internal Hom, as in [15, Theorem 7.1]. Invertibility of D_a , followed by ordinary closed adjunction, gives equivalences of ordinary spectra

$$B_a \simeq F(D_a, D_a) \simeq F(D_a \wedge M_a S, I_{\text{BC}}).$$

The displayed smash is ordinary. Put $C = D_a \wedge M_a S$. Brown–Comenetz representability now gives, naturally in U ,

$$[U, B_a] \cong \text{Hom}_{\mathbb{Z}}(\pi_0(U \wedge C), \mathbb{Q}/\mathbb{Z}_{(p)}).$$

Write U as a filtered homotopy colimit of finite ordinary p -local spectra U_i . Ordinary smash with C and ordinary homotopy groups preserve this colimit, so

$$\pi_0(U \wedge C) = \varinjlim_i \pi_0(U_i \wedge C).$$

The functional corresponding to f vanishes on every term by the assumed finite precompositions. It therefore vanishes on the colimit, and $f = 0$. This argument tests by ordinary finite spectra and uses no interchange of $K(a)$ -localization with this filtered colimit. $\square$

Proof of Proposition 9.1. First obtain some finite exponent for the actual local unit, independently of any finite test. Devinatz–Hopkins prove that every $K(h)$ -local spectrum belongs to the thick tensor ideal generated by E_h in $\mathcal{C}_h$ [7, Appendix A, Proposition A.3]. This is the local nilpotence consequence of Hopkins–Ravenel; its descendability form is [20, Proposition 10.10]. The objects of finite E_h -exponent form a thick tensor ideal containing E_h : modules have exponent at most one, tensoring preserves a bound, and cofiber sequences and retracts preserve finite bounds by Lemma 6.2. It follows that B_h has some finite exponent c . This is one bound on an actual map before any test is chosen; the nilpotence theorem supplies no numerical value needed here.

Apply the full local-unit Amitsur construction in $\mathcal{C}_h$ to B_h , and then the exact ordinary functor $F(V_0, -)$ for a finite ordinary V_0 . Proposition 8.1 identifies its second page with

$$H_{\text{cts}}^s(\mathbb{G}_h, \pi_q(E_h \wedge D V_0)).$$

Proposition 4.3 makes this page zero for $s \geq h^2 + 2 = m$. The tested-image assertion of Proposition 6.3(ii) applies using the already known finite exponent c . It concerns the full finite partial totalizations and their finite fiber cubes, and gives

$$[V_0, I_h^{\otimes h^m}] \longrightarrow [V_0, B_h] \quad \text{zero for every finite ordinary } V_0.$$

Thus the actual arrow $e_m(B_h)$, regarded in ordinary spectra, vanishes after every finite ordinary precomposition. Lemma 9.2 makes it null as an ordinary map and hence null in the full subcategory $\mathcal{C}_h$.

For clarity, all layers needed next occur at finite stages. Set $I_h^{\otimes_h 0} = B_h$, and for $0 \leq j \leq m$ put

$$F_j = \text{cofib}(I_h^{\otimes_h m} \longrightarrow I_h^{\otimes_h(m-j)}).$$

The consecutive maps form a finite filtration $0 = F_0 \rightarrow F_1 \rightarrow \cdots \rightarrow F_m$. The cofiber of $I_h^{\otimes_h(i+1)} \rightarrow I_h^{\otimes_h i}$ is $E_h \otimes_h I_h^{\otimes_h i}$. The cofiber comparison for consecutive composites therefore identifies

$$\text{cofib}(F_j \rightarrow F_{j+1}) \simeq E_h \otimes_h I_h^{\otimes_h(m-j-1)} \quad (0 \leq j < m).$$

Finally $F_m = \text{cofib}(e_m(B_h))$. A nullhomotopy of $e_m(B_h)$ makes B_h a retract of this cofiber. This proves both assertions. $\square$

10 Field-valued points and coherent trace

Continue with $p \geq 5$, $t = p - 1$, $h = p$, and the pure subgroup H of Theorem 7.1. To transport the finite upper layers, we need exponent one for modules over the mixed algebra

$$D = E_t \wedge E_h, \quad D' = L_{K(t)}D.$$

Here H acts on the first factor. Section 11 will make each transformed layer a D' -module in $\mathcal{D}_H$. The $K(t)$ -acyclic spectra form a smash ideal. Monoidal localization of D into $\mathcal{C}_t$, followed by the lax monoidal inclusion into ordinary spectra, therefore gives $D' = L_{K(t)}D$ a commutative algebra structure in $\mathcal{D}_H$. Its localization map is equivariant and is a map of these commutative algebras.

Proposition 10.1 (The completed mixed trace). *The mixed H -algebra D' has an underlying degree-zero class whose H -trace is a unit. Every D' -module in $\mathcal{D}_H$ therefore has H -exponent at most one.*

A trace unit gives the required coherent retraction by induction and coinduction. We first prove that implication. We then construct the trace unit from the two formal-group isomorphisms supplied by the factors of D : the fixed E_h factor forces H to act freely on field-valued points of

$$R = \pi_0 D / \mathfrak{m}_t \pi_0 D.$$

The ideal is H -stable. The mixed cooperation calculation of Lemma 8.2 makes D even periodic, with an orientation and periodic unit from E_t .

Lemma 10.2 (Coherent trace splitting). *Let D' be a commutative algebra in $\mathcal{D}_H$. Suppose an underlying class $z \in \pi_0 D'$ has unit trace*

$$v = \sum_{g \in H} g(z) \in (\pi_0 D')^\times.$$

Then every D' -module U in $\mathcal{D}_H$ has $\exp_H(U) \leq 1$.

Proof. Since H is finite, the underlying object $A_H \wedge U$ is $F(H_+, U)$, with diagonal action. Untwisting the value at g by the action of g^{-1} on U identifies it with the coinduction of the underlying spectrum:

$$\theta : A_H \wedge U \xrightarrow{\simeq} \text{Coind}_1^H(\text{Res}_1^H U).$$

Indeed, on functions the diagonal action is $(hf)(g) = hf(h^{-1}g)$, and $(\theta f)(g) = g^{-1}f(g)$ turns it into translation $(h\theta f)(g) = (\theta f)(h^{-1}g)$. For finite H , the finite coproduct-product equivalence in spectra identifies induction with coinduction. Induction adjunction takes the underlying multiplication map $\mu_z : U \rightarrow U$ to a map with coherent H -action

$$r_z : A_H \wedge U \longrightarrow U.$$

Let $j : U \rightarrow A_H \wedge U$ be the unit map. Under θ , its component at g is $g^{-1} : U \rightarrow U$. The induction map has component $g\mu_z$ at g , so the underlying composite is

$$r_z j = \sum_{g \in H} g\mu_z g^{-1} = \mu_{\sum_g g(z)} = \mu_v.$$

The middle equality uses the compatibility of the D' -module structure with the H -action. Since v is a unit, this is an equivalence of underlying spectra. Equivalences of H -diagrams are detected on underlying objects, so $r_z j$ is an equivalence in $\mathcal{D}_H$. Hence $(r_z j)^{-1} r_z$ retracts j there. Its fiber map is null, which is exponent at most one. The class z was only an underlying class; no homotopy-fixed lift of it is needed. $\square$

Lemma 10.3 (Freeness on field-valued points). *For every unital ring map $x : R \rightarrow \Omega$ to a field, the equality $x \circ g = x$ for $g \in H$ implies $g = 1$.*

Proof. We first recall explicitly the naturality of the formal group used here. For a complex-oriented even periodic commutative ring spectrum B , an orientation and a periodic unit give a coordinate in degree zero. The finite projective-space calculation and its surjective restriction tower give

$$B^0(\mathbb{CP}^\infty) = (\pi_0 B)[[z_B]].$$

Products of projective spaces give the corresponding group law. A unital ring-spectrum map $f : B \rightarrow C$ carries an orientation to an orientation and a periodic unit to a unit. Its coordinate therefore has an invertible linear coefficient in the chosen coordinate of C . The inverse of this coordinate substitution gives an isomorphism from the base change of the formal group of B to the formal group of C . The product calculation gives compatibility with group laws, and this construction respects composites of ring-spectrum maps. All later base changes are base changes of these algebraic formal groups along maps of coefficient rings.

Let $\rho : \pi_0 D \rightarrow \Omega$ be the composite through x . Write $\iota_t : E_t \rightarrow D$ and $\iota_h : E_h \rightarrow D$ for the two factor maps, and let ρ_t, ρ_h be their coefficient composites with ρ . The map ρ_t factors through

$$O_t/\mathfrak{m}_t = k_t \longrightarrow \Omega.$$

The last map is injective, since it is a unital map from a field. After base change along ρ_t, ρ_h, ρ , the factor maps give formal-group isomorphisms

$$\phi_t : \mathcal{F}_{t,\Omega} \xrightarrow{\cong} \mathcal{F}_{D,\Omega}, \quad \phi_h : \mathcal{F}_{h,\Omega} \xrightarrow{\cong} \mathcal{F}_{D,\Omega}.$$

Let g_D denote the action of g on D . The equality $x \circ g = x$ implies $\rho \circ \pi_0(g_D) = \rho$, so naturality makes g_D an automorphism u_g of the single formal group $\mathcal{F}_{D,\Omega}$. Since $g_D \iota_h = \iota_h$, naturality with the second factor gives $u_g \phi_h = \phi_h$. Thus u_g is the identity. But $g_D \iota_t = \iota_t g$; naturality with the first factor gives

$$u_g \phi_t = \phi_t g_\Omega,$$

where g_Ω is the specialization of the prescribed stabilizer automorphism. Consequently g_Ω is the identity.

On the special fiber the pure stabilizer element g is its specified automorphism of the Honda formal group over k_t . This is the special-fiber content of Lubin–Tate deformation [19, Proposition 3.3 and paragraph 3.4], with its coherent realization as above. If $g \neq 1$, some coefficient of the power series $g(X) - X$ is nonzero in k_t . It remains nonzero under the injective map $k_t \rightarrow \Omega$, contrary to $g_\Omega = 1$. Therefore $g = 1$.

This proof used only the coefficient map to Ω for algebraic base change; it did not require a ring-spectrum map to a spectrum with coefficient field Ω . The assertion concerns equality of actual field-valued maps; invariance of their kernels alone would not give the conclusion needed next. $\square$

Lemma 10.4 (Algebraic trace one). *Let a finite group H act on a commutative ring R . Suppose that for every unital field-valued map $x : R \rightarrow \Omega$, $x \circ g = x$ implies $g = 1$. Then*

$$1 \in \text{im}(\text{Tr} : R \rightarrow R^H), \quad \text{Tr}(r) = \sum_{g \in H} g(r).$$

Proof. The trace image is an ideal of R^H , since trace is additive and $a \text{Tr}(r) = \text{Tr}(ar)$ for $a \in R^H$. Suppose this ideal is proper, and choose a maximal ideal $\mathfrak{q}$ containing it. For every $r \in R$, the monic orbit polynomial $\prod_{g \in H} (X - g(r))$ has coefficients in R^H . Thus R is integral over R^H . Lying over gives a prime $\mathfrak{p}$ of R above $\mathfrak{q}$. The quotient and inclusion into its fraction field give a unital map $x : R \rightarrow \Omega = \text{Frac}(R/\mathfrak{p})$ killing $\mathfrak{q}$.

The multiplicative unital maps $\chi_g : R \rightarrow \Omega$, $\chi_g(r) = x(g(r))$, are pairwise distinct by the hypothesis. Such distinct maps are linearly independent as functions. To prove this, suppose $\sum_{i=1}^v a_i \chi_i = 0$ is a nonzero relation with the least possible number of nonzero coefficients. One term is impossible by evaluation at 1. If $v > 1$, choose an element $b \in R$ distinguishing χ_1 from χ_v . Subtract $\chi_v(b)$ times the relation at r from the relation at br . The result is

$$\sum_{i < v} a_i (\chi_i(b) - \chi_v(b)) \chi_i(r) = 0.$$

Its coefficient at $i = 1$ is nonzero in the field, giving a shorter nonzero relation, a contradiction.

In particular the function $\sum_{g \in H} \chi_g$ is not zero: each of its coefficients is 1 in the field, even if the characteristic divides $|H|$. On the other hand it equals $x \circ \text{Tr}$, which is zero since x kills the trace ideal. This contradiction proves the claim. If R is the zero ring, the conclusion holds directly with $1 = 0$. $\square$

Proof of Proposition 10.1. Lemmas 10.3 and 10.4 give $\bar{z} \in R$ with $\text{Tr}(\bar{z}) = 1$. Lift it to $z_0 \in \pi_0 D$, and put $v = \sum_g g(z_0)$. Then

$$v - 1 \in \mathfrak{m}_t \pi_0 D.$$

The mixed module $\pi_* D$ is flat over $(E_t)_*$ by Lemma 8.2. Lemma 8.4 therefore identifies the homotopy of its actual localization $D \rightarrow D'$ with the ordinary $\mathfrak{m}_t$ -adic completion of $\pi_* D$.

At the finite adic stage $\pi_0 D / \mathfrak{m}_t^j \pi_0 D$, the element

$$w_j = \sum_{k=0}^{j-1} (1-v)^k$$

satisfies $vw_j = 1 - (1-v)^j = 1$. These inverses are compatible under reduction. Here j is a finite truncation level; the quotient rings themselves are not required to be finite sets. The comparison

with completion is induced by the actual ring map $D \rightarrow D'$. Multiplication by the image of v is O_t -linear and adically continuous and agrees with multiplication by v on the dense ordinary image. It hence agrees with the levelwise multiplication on the completion. The compatible w_j give an inverse to the image of v in $\pi_0 D'$.

Localization is equivariant, so that image of v is the H -trace of the image of z_0 . Lemma 10.2 now applies. If the completed ring is zero, its unit is zero and every unital module is contractible, giving the same conclusion. $\square$

11 Transport of the finite layers and the canonical unit

The upper null map supplies a retract built from m local E_h -module layers. We now transport those individual layers by the exact functor on ordinary p -local spectra

$$T : \mathrm{Sp}_{(p)} \longrightarrow \mathcal{D}_H, \quad T(V) = L_{K(t)}(E_t \wedge V),$$

with H acting through the coherent action on E_t . Applying T to a local layer means applying it to the underlying ordinary spectrum of that object of $\mathcal{C}_h$. The mixed trace will make each resulting layer have H -exponent at most one.

Proposition 11.1 (The cross-height upper exponent). *Let $p \geq 5$ be prime, with $t = p - 1$ and $h = p$. One has*

$$\exp_H(T(B_h)) \leq m = p^2 + 2.$$

Proof. The inclusion $\mathcal{C}_h \rightarrow \mathrm{Sp}_{(p)}$ is exact and lax symmetric monoidal. Since E_h is local, the localization unit identifies the included local algebra E_h with its given ordinary algebra. Each local layer $E_h \otimes_h I_h^{\otimes h^i}$ in Proposition 9.1 consequently has an ordinary E_h -module structure. Ordinary smash with E_t makes it a $D = E_t \wedge E_h$ -module with compatible H -action.

The $K(t)$ -acyclic spectra form a smash ideal. Thus the localization into $\mathcal{C}_t$ is symmetric monoidal for its localized tensor, and the inclusion back into ordinary spectra is lax symmetric monoidal. Applying localization to the algebra and this module gives, after that inclusion, a D' -module structure on the transformed layer in $\mathcal{D}_H$. Proposition 10.1 gives every such layer H -exponent at most one.

The functor T preserves cofiber sequences and retracts. Apply it to the finite filtration and retraction of Proposition 9.1, and add the m layer bounds using Lemma 6.2. This yields the stated bound. The construction used ordinary module structures on the individual layers; it required no preservation by T of the $\mathcal{C}_h$ tensor product. $\square$

Corollary 11.2 (The exponent contradiction for the canonical unit). *Let $p \geq 5$ be prime, put $t = p - 1$ and $h = p$, and let $X = S_p^\wedge$ be the derived p -completed sphere. The canonical map*

$$i_{h,X} = L_t(\eta_X^{(h)}) : L_t X \longrightarrow L_t L_{K(h)} X$$

has no homotopy retraction in ordinary spectra.

Proof. The functor T inverts $K(t)$ -equivalences: the cofiber of such a map is $K(t)$ -acyclic, and remains so after ordinary smash with E_t . Every L_t -localization unit is a $K(t)$ -equivalence. Lemma 2.1 says that the actual completion $c : S \rightarrow X$ is a $K(a)$ -equivalence for each positive a , in particular

for $a = t, h$. Write $\lambda_Z : Z \rightarrow L_t Z$ for the lower localization unit. Consider the square of actual localization and completion maps

$$\begin{array}{ccc} S & \xrightarrow{\eta_S^{(h)}} & B_h \\ \lambda_X c \downarrow & & \downarrow \lambda_{L_{K(h)} X} L_{K(h)}(c) \\ L_t X & \xrightarrow{i_{h,X}} & L_t L_{K(h)} X. \end{array}$$

It commutes by naturality of the two localization units. Both vertical arrows become equivalences under T : for the left one use the $K(t)$ -equivalences just stated, and for the right one use that $L_{K(h)}(c)$ is an equivalence followed by a $K(t)$ -equivalence. Moreover $T(S) \simeq E_t$, with coherent H -action, since E_t is $K(t)$ -local.

A homotopy retraction of $i_{h,X}$ would therefore, after applying T and these identifications, exhibit E_t as a retract of $T(B_h)$ in $\mathcal{D}_H$. Proposition 11.1 and preservation of exponent bounds under retracts would give $\exp_H(E_t) \leq m$. The finite-page lower result Proposition 7.2 gives the strict inequality $\exp_H(E_t) > m$, a contradiction. $\square$

Proof of Theorem 1.4. Proposition 7.2 gives the strict lower bound and Proposition 11.1 gives the upper bound, both for $m = p^2 + 2$ in the single category $\mathcal{D}_H$. The preceding corollary proves that a retraction of the canonical map would turn the lower object into an H -equivariant retract of the upper object and contradict those bounds. This proves every assertion of the theorem. The lower bound uses finite-page survival only; the upper bound uses the actual null map and its finite layers. $\square$

A Ordinary products in homotopy fixed points

We prove Lemma 3.2 by constructing relative products on the ordinary bar tower. The construction supplies cup products and differential derivations independently of convergence; separated convergence identifies the final product with the product on actual homotopy groups.

Proof of Lemma 3.2. Use the homogeneous free bar construction $EQ_\bullet$, with $EQ_q = Q^{q+1}$, and let $EQ^{(s)}$ be its s -skeleton. The full totalization tower of the group-action resolution is

$$V_s = F_Q((EQ^{(s)})_+, R), \quad \text{holim}_s V_s = F_Q(EQ_+, R) = R^{hQ}.$$

Here F_Q denotes the derived spectrum of equivariant maps. The skeletal filtration gives the usual homogeneous cochains and hence the displayed E_2 page.

For the prismatic geometry below, compare Basterra–Mandell [2, proof of Theorem 7.1, pp. 23–26]. Their bar diagonal for a (partial) non-unital E_1 algebra over a base ring spectrum H uses the same half-cut regions and inverse barycentric coordinates. Here we construct the equivariant relative maps for EQ and identify the resulting E_∞ product with the associated-graded product on actual homotopy groups under the lemma’s stated filtration hypothesis. For barycentric coordinates $t = (t_0, \dots, t_q) \in \Delta^q$, set $c_{-1} = 0$ and $c_i = t_0 + \dots + t_i$. Define two barycentric vectors by

$$\begin{aligned} \ell_i(t) &= 2(\min(c_i, \tfrac{1}{2}) - \min(c_{i-1}, \tfrac{1}{2})), \\ \rho_i(t) &= 2(\max(c_i, \tfrac{1}{2}) - \max(c_{i-1}, \tfrac{1}{2})). \end{aligned} \tag{22}$$

They are nonnegative and each has sum one. Equivalently, their i th coordinates are twice the lengths of the intersections of $[c_{i-1}, c_i]$ with $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. The formulas are continuous, including when a cumulative sum equals $\frac{1}{2}$.

For an order-preserving map $\theta : [q] \rightarrow [q']$, let $(\theta_* t)_j = \sum_{\theta(i)=j} t_i$. Each interval belonging to $\theta_* t$ is the union of the consecutive intervals belonging to the corresponding fiber of θ ; an empty fiber gives an interval of length zero. Additivity of intersection lengths gives

$$\ell(\theta_* t) = \theta_* \ell(t), \quad \rho(\theta_* t) = \theta_* \rho(t).$$

The realization relation is $[\theta^* \sigma, t] = [\sigma, \theta_* t]$ for $\sigma \in EQ_{q'}$. Thus for every bar simplex $\sigma \in EQ_q$ the formula

$$\delta[\sigma, t] = ([\sigma, \ell(t)], [\sigma, \rho(t)])$$

is compatible with all faces and degeneracies and defines a continuous map $\delta : EQ \rightarrow EQ \times EQ$. It is Q -equivariant because the action changes σ and leaves the barycentric coordinates fixed. Moreover

$$[\sigma, t] \mapsto ([\sigma, (1 - \tau)t + \tau \ell(t)], [\sigma, (1 - \tau)t + \tau \rho(t)]), \quad 0 \leq \tau \leq 1,$$

is compatible with the same identifications and is equivariant. It is a homotopy from the ordinary diagonal to δ .

Choose k with $c_{k-1} \leq \frac{1}{2} \leq c_k$. The support of $\ell(t)$ is contained in the front face $[0, \dots, k]$, and the support of $\rho(t)$ is contained in the back face $[k, \dots, q]$. These faces have dimensions k and $q - k$. Degenerate faces can only lower those dimensions. Consequently

$$\delta(EQ^{(s)}) \subset \bigcup_{i+j \leq s} EQ^{(i)} \times EQ^{(j)}.$$

This also identifies the cellular chain map. On the region $c_{k-1} \leq \frac{1}{2} \leq c_k$, the map $t \mapsto (\ell(t), \rho(t))$ is a homeomorphism onto $\Delta^k \times \Delta^{q-k}$, with inverse

$$t_i = \ell_i/2 \ (i < k), \quad t_k = (\ell_k + \rho_k)/2, \quad t_i = \rho_i/2 \ (i > k).$$

In the oriented coordinates $t_1, \dots, t_q$ on the source and $\ell_1, \dots, \ell_k, \rho_{k+1}, \dots, \rho_q$ on the product, its Jacobian determinant is $2^q > 0$. The regions meet on their boundaries. The image of the cellular fundamental chain is therefore

$$[\sigma] \mapsto \sum_{k=0}^q [\sigma|_{[0, \dots, k]}] \otimes [\sigma|_{[k, \dots, q]}],$$

with degenerate terms omitted in normalized chains. This is the Alexander–Whitney diagonal, now obtained from the explicit map.

Write $EQ/EQ^{(s)}$ for the based cofiber of $(EQ^{(s)})_+ \rightarrow EQ_+$, and set $EQ^{(-1)} = \emptyset$. The skeletal inclusions are Q -CW cofibrations, so these quotients represent their homotopy cofibers. For $a, b \geq 0$, the skeletal containment above makes the composite of δ with the quotient to $(EQ/EQ^{(a-1)}) \wedge (EQ/EQ^{(b-1)})$ constant on $EQ^{(a+b-1)}$: if $i + j < a + b$, then $i < a$ or $j < b$. We get a Q -equivariant relative map

$$EQ/EQ^{(a+b-1)} \longrightarrow (EQ/EQ^{(a-1)}) \wedge (EQ/EQ^{(b-1)}).$$

Multiplication on R and precomposition give

$$F_Q(EQ/EQ^{(a-1)}, R) \wedge F_Q(EQ/EQ^{(b-1)}, R) \longrightarrow F_Q(EQ/EQ^{(a+b-1)}, R).$$

These maps are compatible as a and b vary, since they are induced by the same δ and the nested skeletal quotients.

For completeness, the associativity needed for powers also holds within this relative filtration. Partition $[0, 1]$ into any m successive intervals of positive lengths. For the j th output vector, divide the lengths of its intersections with $[c_{i-1}, c_i]$ by the length of the j th interval. The preceding naturality and equivariance proof applies verbatim. If k_j is an index whose coordinate interval contains the j th cut, the supports lie in the successive faces $[0, \dots, k_1], [k_1, \dots, k_2], \dots, [k_{m-1}, \dots, q]$. Their dimensions sum to q ; boundary cases can only reduce the sum. Thus the same construction gives m -fold relative maps. Composing cut maps is exactly refinement of the partition. The positive interval lengths form a convex simplex, so the partitions for two parenthesizations are joined by varying those lengths. This gives a homotopy through maps with the same relative skeletal property. Thus the relative products are associative up to filtered homotopy, as required for their iterated products.

Put $I_s = F_Q(EQ/EQ^{(s-1)}, R)$ for $s \geq 0$. The quotient cofiber sequence for two successive skeleta gives

$$\text{cofib}(I_{s+1} \rightarrow I_s) \simeq F_Q(EQ^{(s)}/EQ^{(s-1)}, R) \simeq \text{fib}(V_s \rightarrow V_{s-1}).$$

The fiber sequences $I_s \rightarrow R^{hQ} \rightarrow V_{s-1}$ form a diagram with the identity on R^{hQ} and the tower transition on the right. The stable fiber diagram identifies the connecting maps of the displayed cofiber sequences with those of the V_s tower, with the usual rotation signs. Thus the paired relative spectral sequence is the ordinary tower spectral sequence with the same filtration index. The relative products pair these sequences and their exact couples. On cellular cochains the displayed chain formula is the ordinary cup product; in inhomogeneous coordinates its value is

$$(v \smile w)(g_1, \dots, g_{a+b}) = v(g_1, \dots, g_a) (g_1 \cdots g_a) w(g_{a+1}, \dots, g_{a+b}),$$

with the usual graded signs. The cellular boundary formula for a product, together with the internal Koszul sign, gives the derivation rule in the paired exact couples and hence on every spectral-sequence page. This E_2 product is the ordinary cup product used by the ordinary-to-Tate comparison in Section 7. Since each subsequent product is induced on homology, the products agree on every later page as well.

The fiber sequence for I_a identifies its image in $\pi_* R^{hQ}$ with $\ker(\pi_* R^{hQ} \rightarrow \pi_* V_{a-1})$. After forgetting the relative conditions, the product just constructed is homotopic to the actual product on $F_Q(EQ_+, R)$ by the equivariant homotopy from δ to the diagonal. Thus these relative products multiply the actual filtration groups. Passing to successive relative quotients is exactly the product in the paired spectral sequence. When the stated convergence identifies those quotients with E_∞ , this is the associated-graded homotopy product as stated. $\square$

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